



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**



A DEEP LEARNING APPROACH FOR COLORIZING BLACK-AND-WHITE PHOTOGRAPHS *in Vietnam before 1950*

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CHAPTER 01: INTRODUCTION



1.1: PROBLEM STATEMENT

- Black-and-white photographs are valuable historical records, but the lack of color diminishes their emotional impact and visual meaning.
- Color helps distinguish objects, connect emotions, and enhance the viewer's interaction with the image.
- Most historical photographs in Vietnam before 1950 are black-and-white, making it difficult for younger generations to relate to them.
- Colorizing black-and-white photos is a technical challenge, as one pixel can correspond to multiple colors, making accurate colorization difficult.





1.2: RESEARCH OBJECTIVES

- Develop a deep learning-based model for colorizing historical black-and-white photos from Vietnam before 1950.
- Deploy the model as a user-friendly web application for easy colorization of grayscale images.

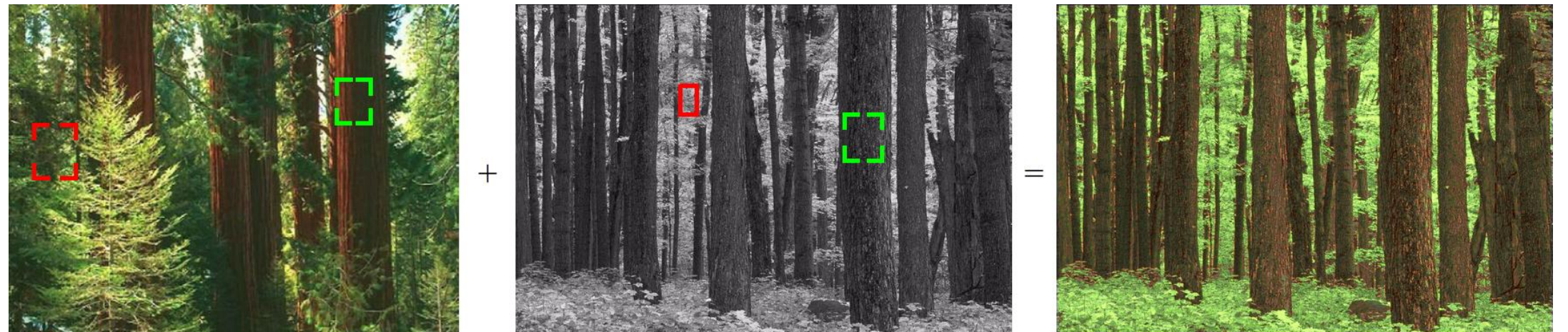
1.3. LITERATURE REVIEW (1)

01 Rule-based methods

- Early techniques focused on manual and rule-based approaches for image colorization.
- Limitations: Struggled with handling complex images and achieving realistic results.



Levin et al. in SIGGRAPH 2004

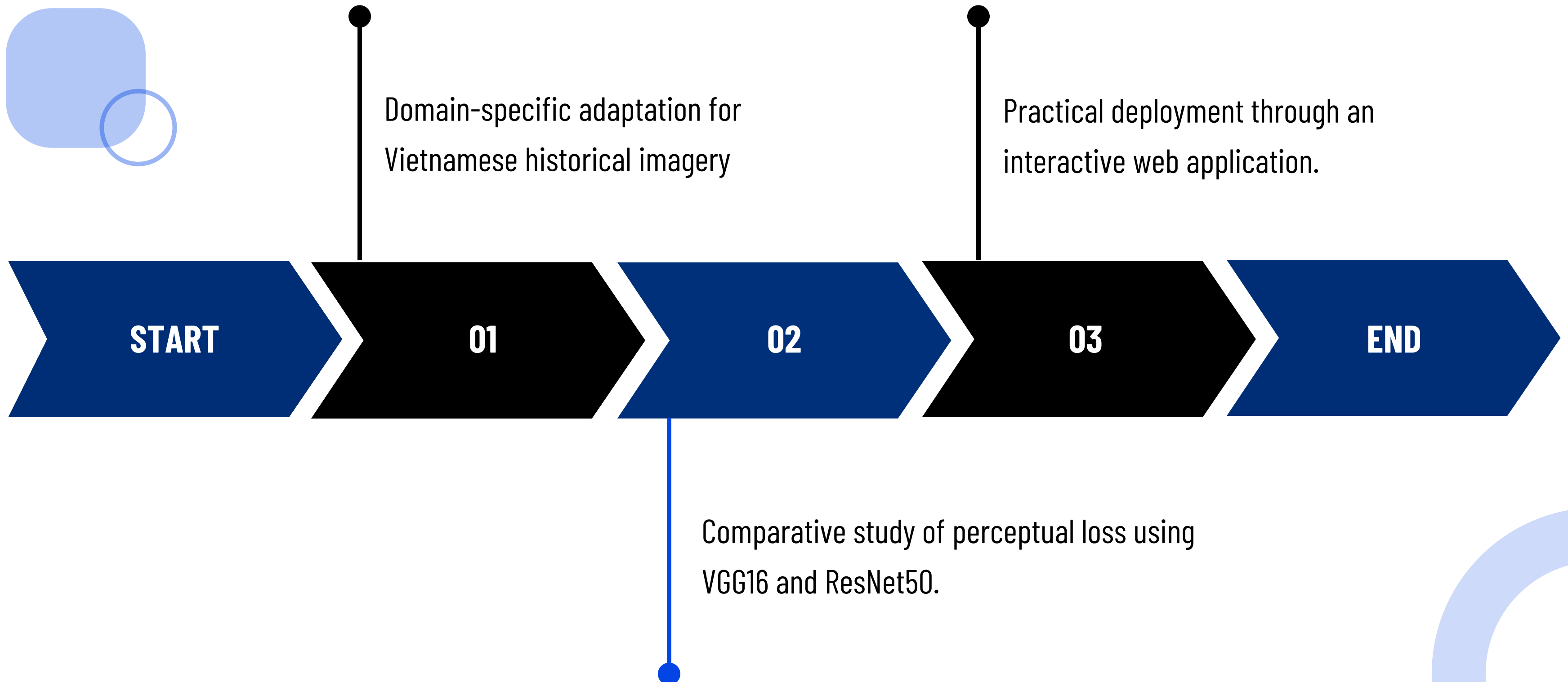


Welsh et al. in SIGGRAPH 2002

02 Deep Learning-based methods



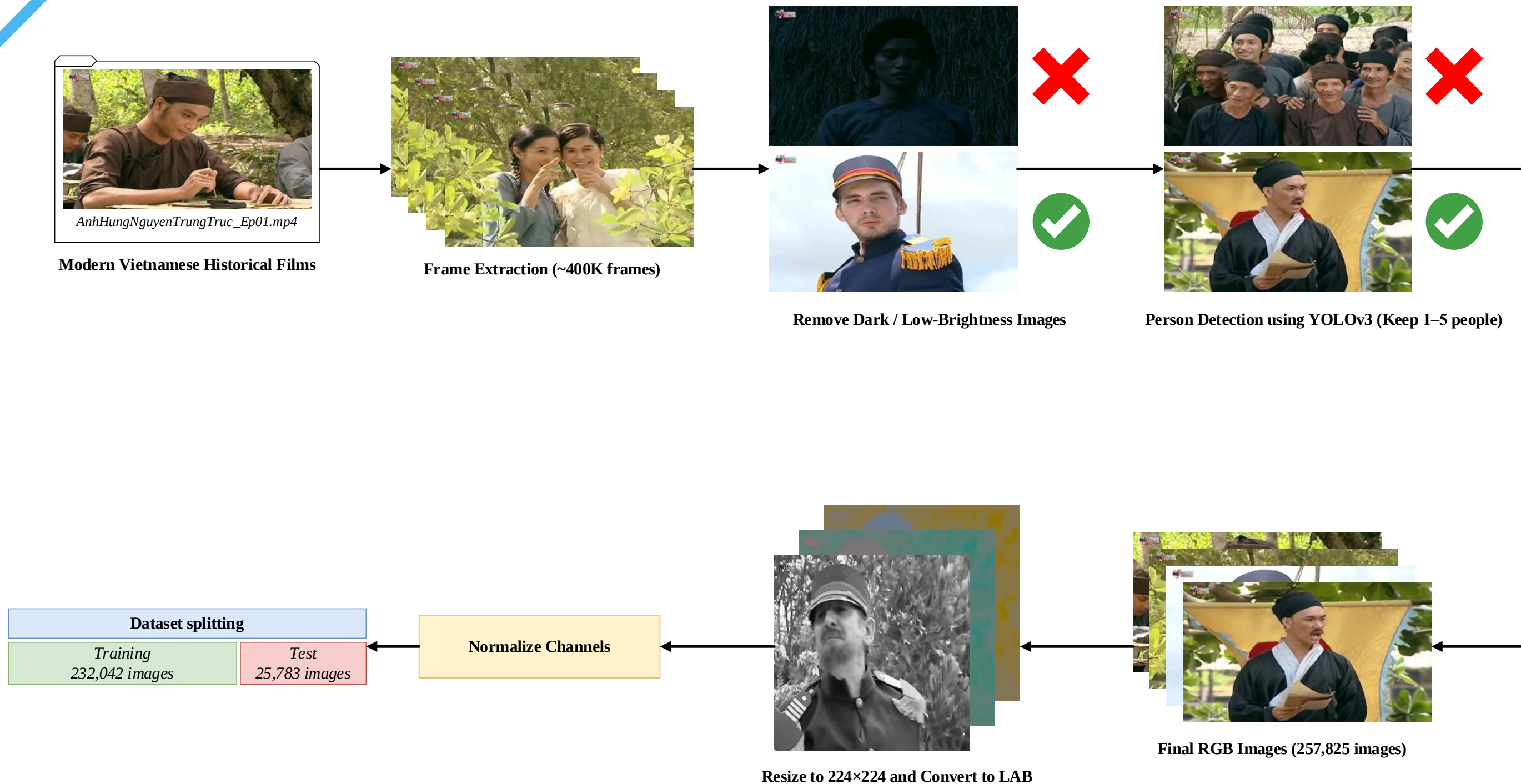
1.4. THESIS CONTRIBUTIONS





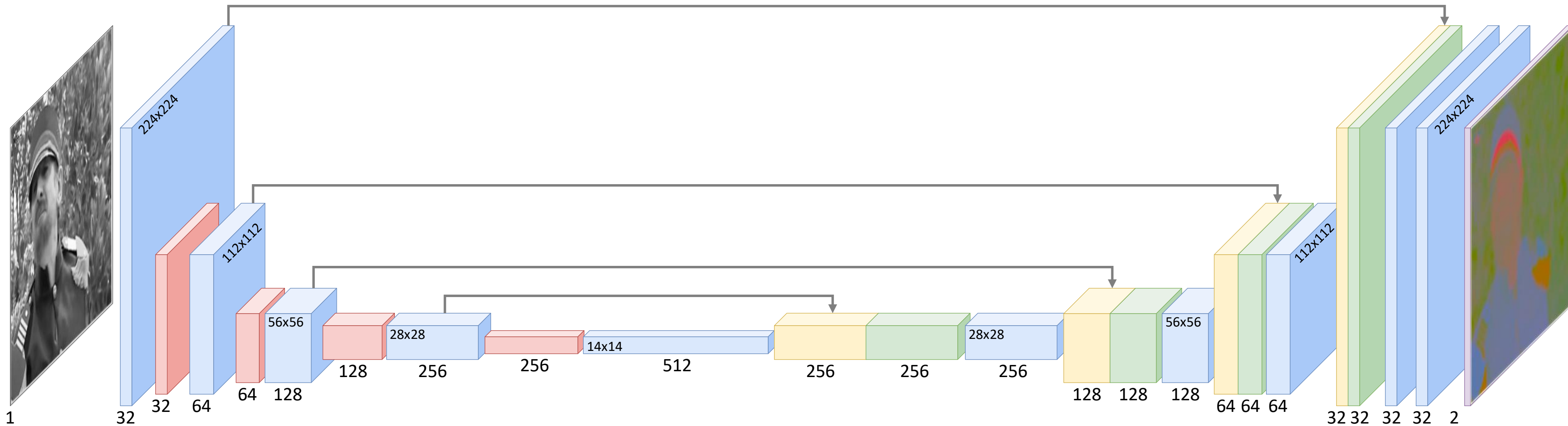
CHAPTER 2: METHODOLOGY

2.1. DATASET PREPARATION

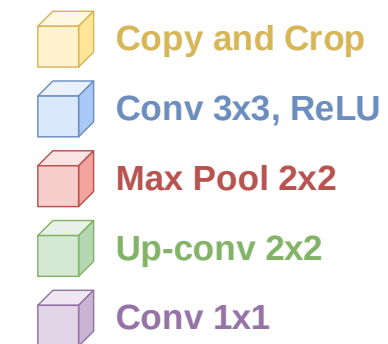


1. Data Source: Extracted frames from modern Vietnamese historical films
2. Removed dark, blurry, or irrelevant frames.
3. Used YOLOv3 to retain frames with 1-5 people.
4. 257,825 clean RGB images, converted to LAB and split into training and test sets.

2.2. MODEL ARCHITECTURE (U-NET)



- U-Net with encoder-decoder structure and skip connections for efficient feature reuse.
- Input and Output: Grayscale L channel input, predicts missing A and B chrominance channels.
- Network Structure: Conv2D, MaxPooling (downsampling), Conv2DTranspose (upsampling), skip connections.

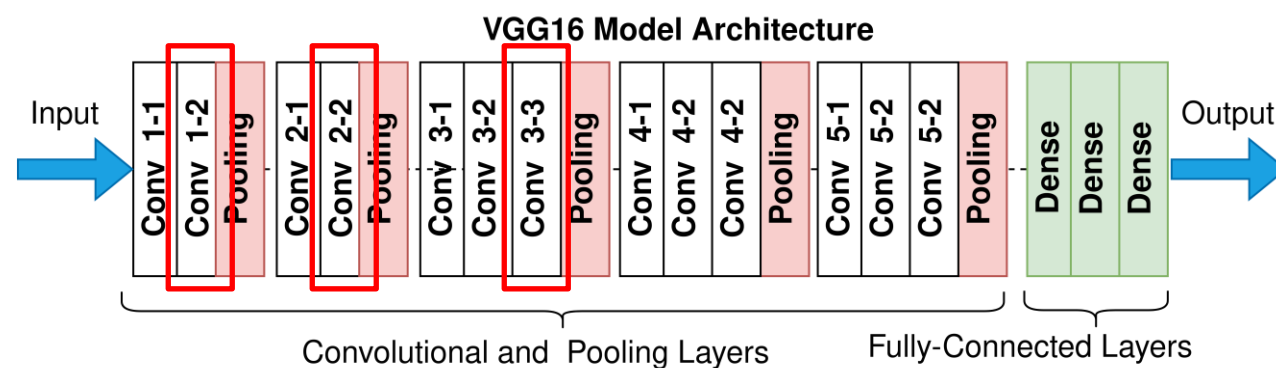


2.3. LOSS FUNCTIONS

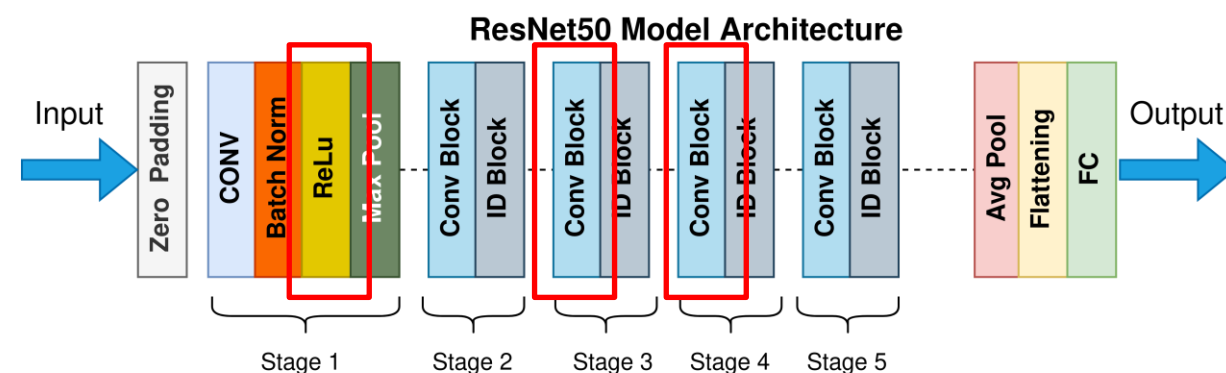
$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (y_i^{\text{true}} - y_i^{\text{pred}})^2$$

- MSE Loss: Computes pixel-level differences.

$$\mathcal{L}_{\text{perceptual}} = \sum_{l=1}^L \frac{1}{N_l} \sum_{i=1}^{N_l} \left(F_l^{\text{true}}(i) - F_l^{\text{pred}}(i) \right)^2$$



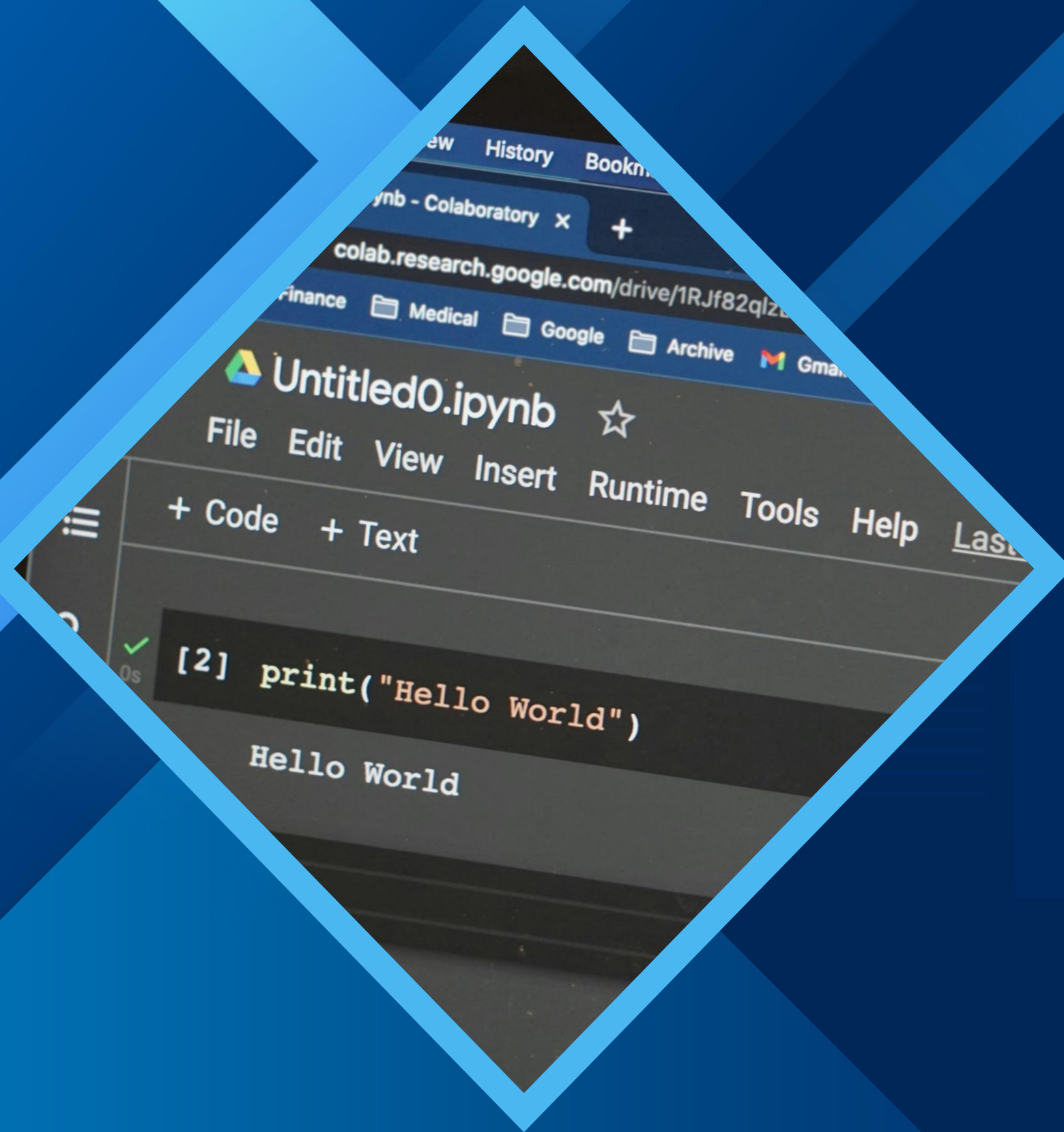
- VGG16 Perceptual Loss: Uses features extracted from VGG to assess perceptual quality.



- ResNet50 Perceptual Loss: Offers deeper feature representation, focusing on high-level semantics.

2.4. TRAINING SETUP

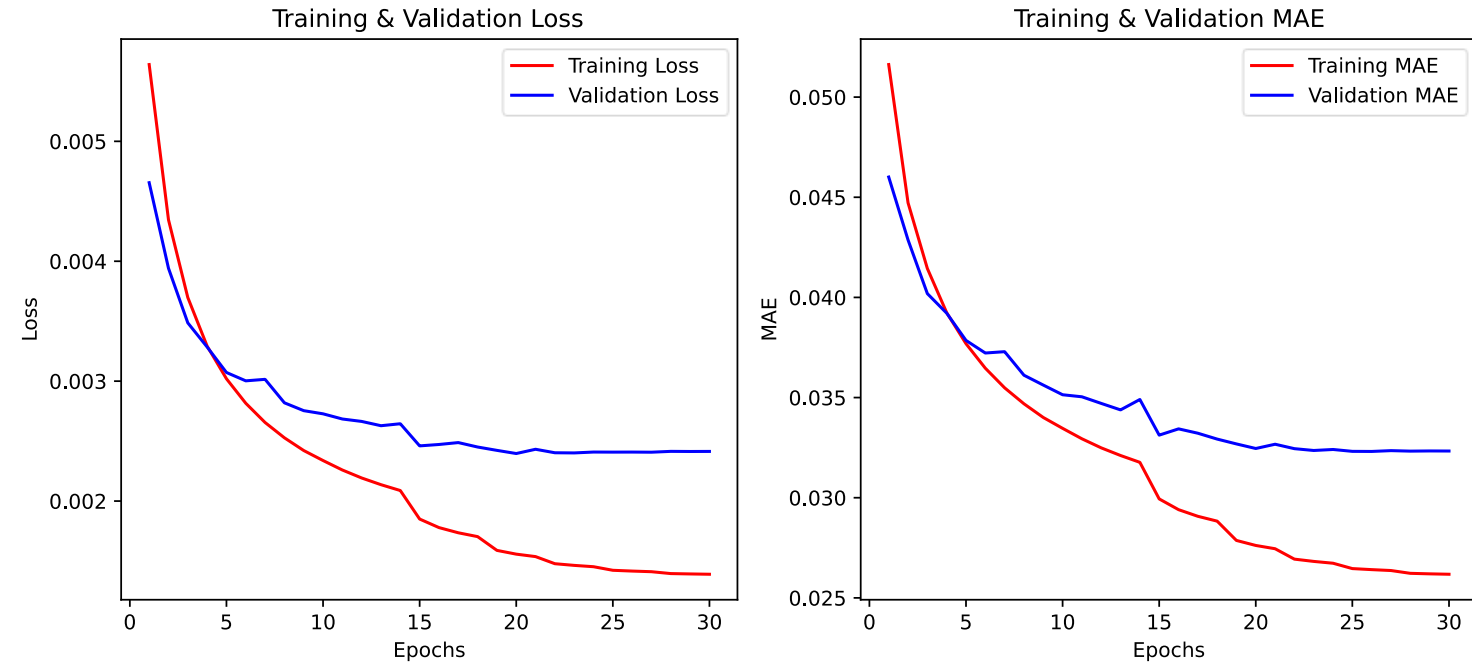
- Three U-Net-based models were trained using different loss functions: MSE, Perceptual Loss using VGG16, and Perceptual Loss using ResNet50.
- The models were trained for 30 epochs with a batch size of 16, using a dataset of 232,042 preprocessed images.
- Training was conducted on a cloud-based NVIDIA RTX 3090 GPU to ensure stable performance.



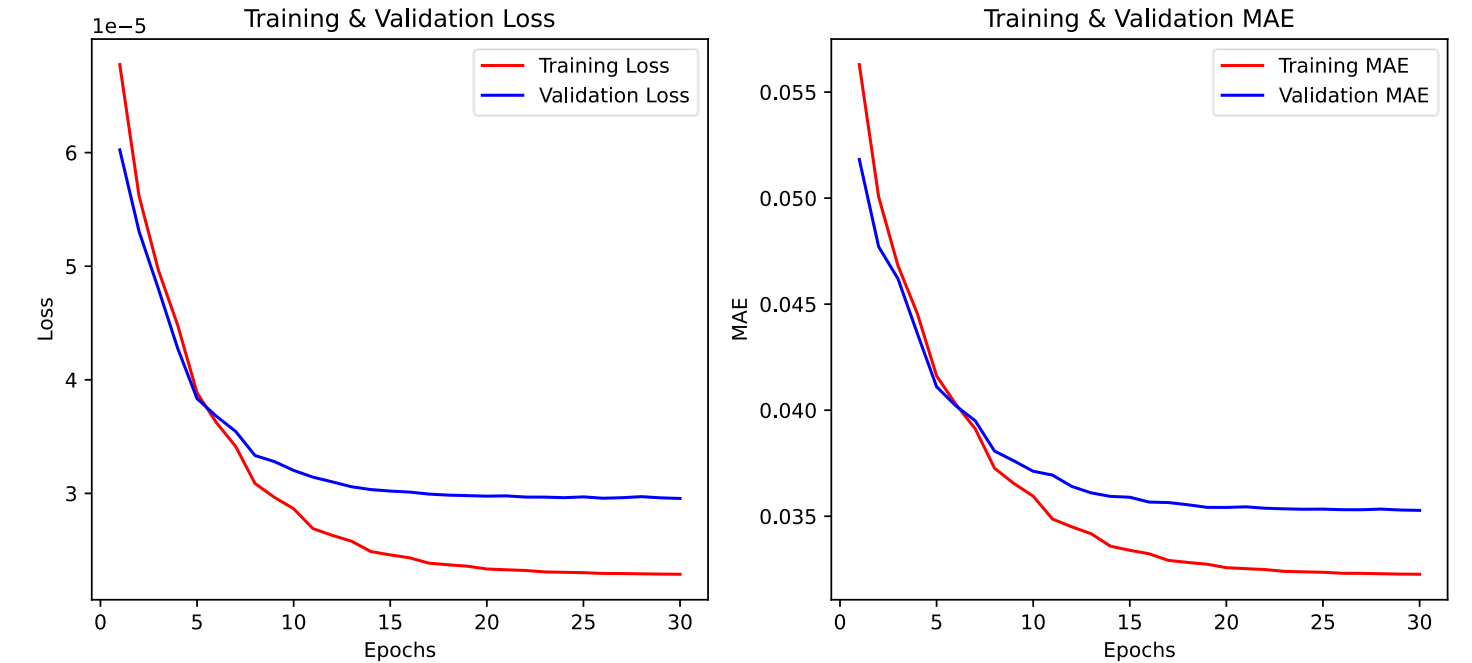
CHAPTER 03: EXPERIMENTS & RESULTS



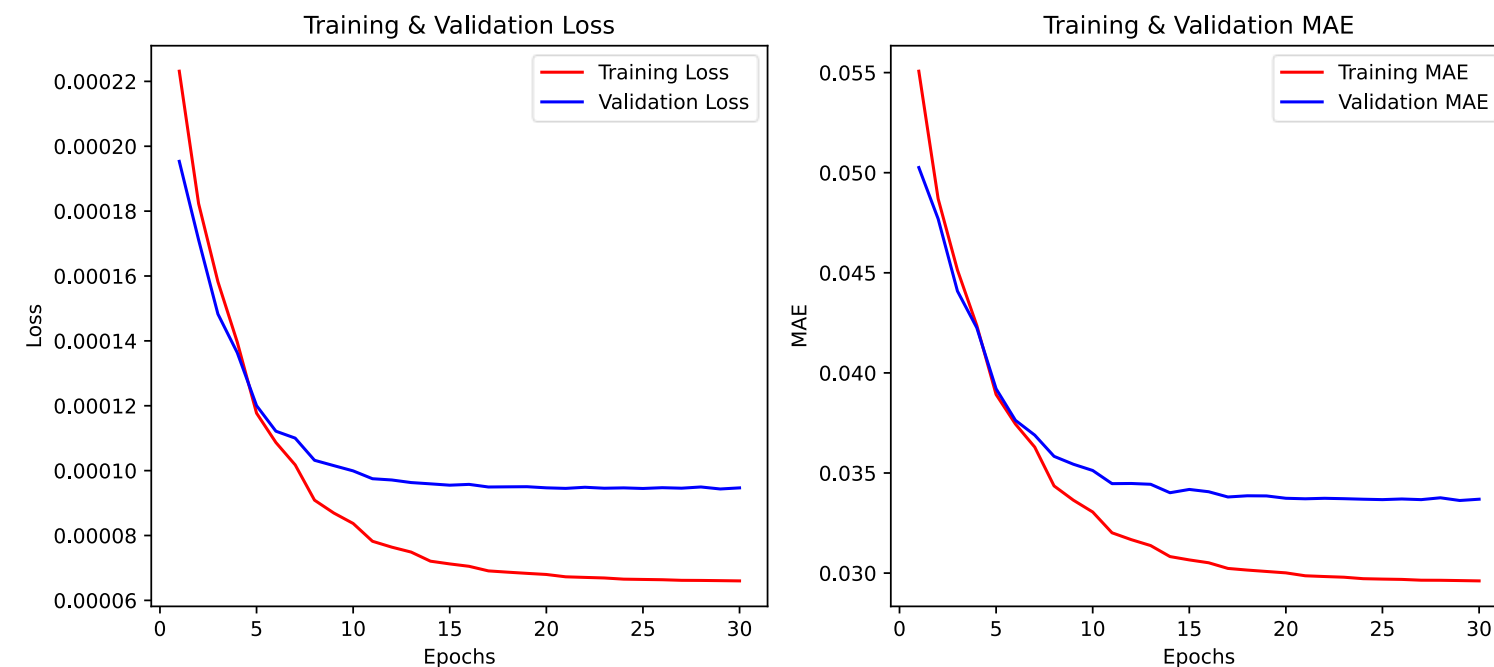
3.1. TRAINING CURVES



MSE



Perceptual Loss using VGG16



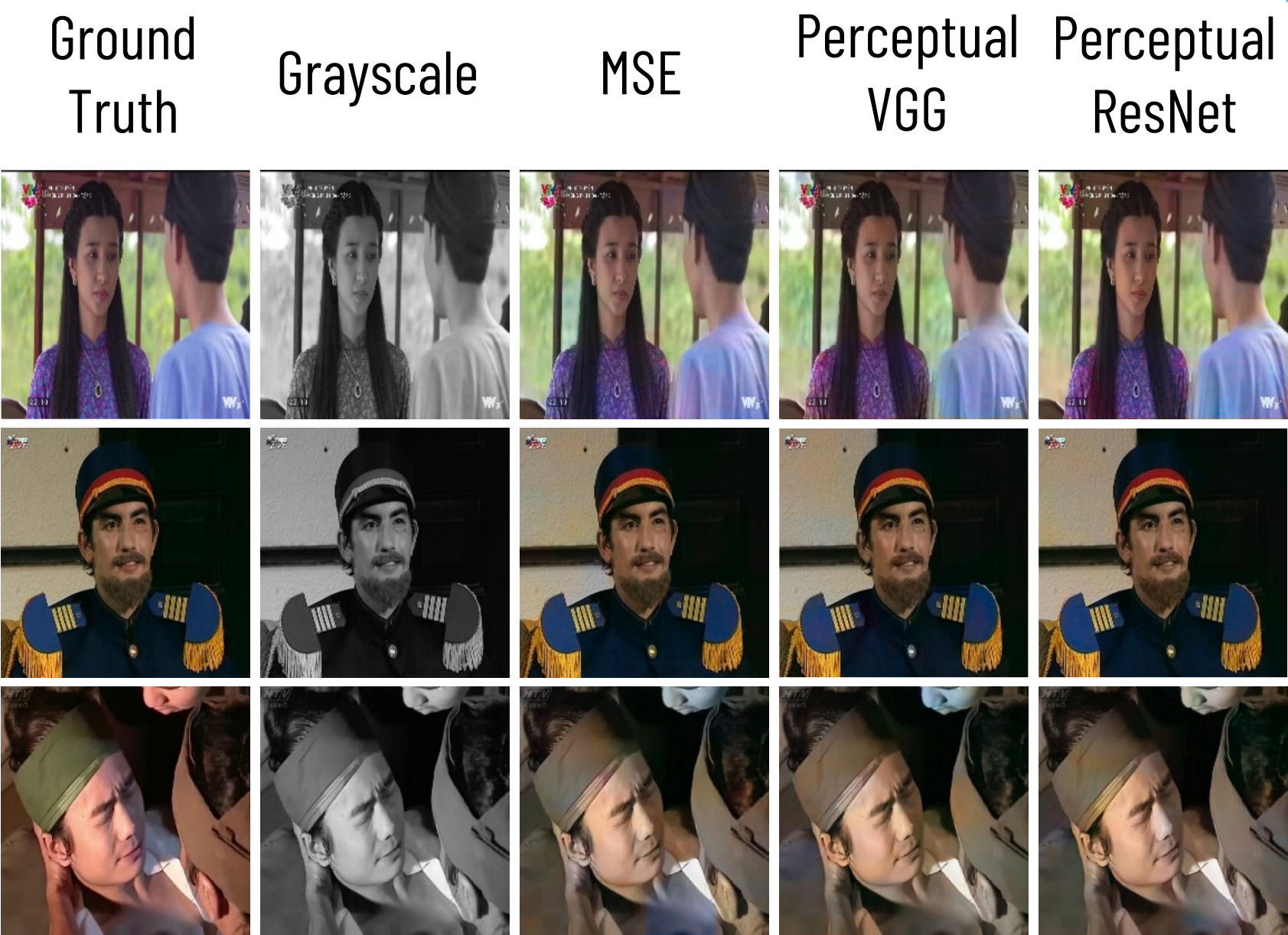
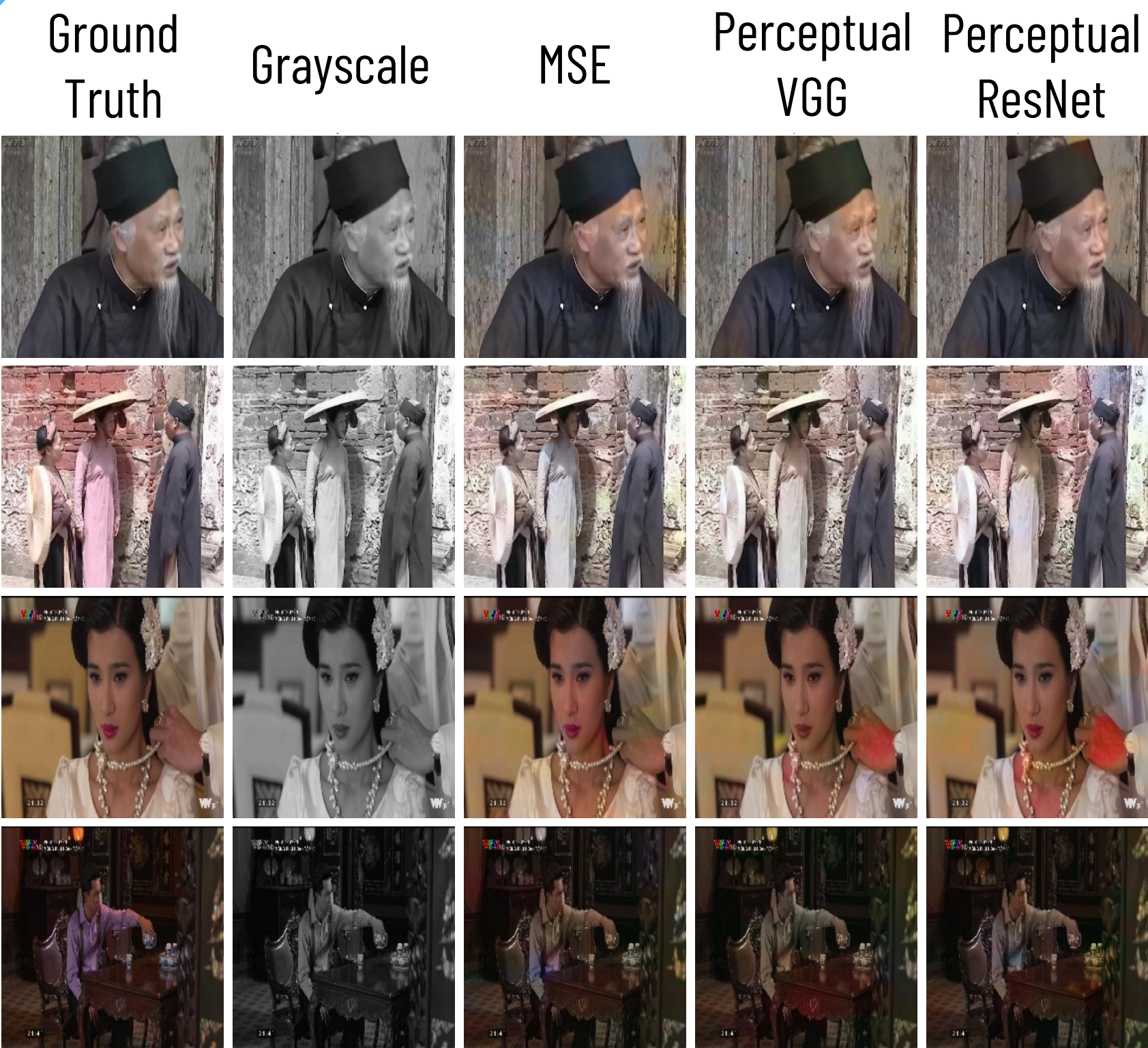
Perceptual Loss using ResNet50

3.2. QUANTITATIVE RESULTS

- Evaluated using MAE, Peak Signal to Noise Ratio and Structural Similarity Index.
- MSE had higher PSNR but lower perceptual quality.
- VGG16-based model achieved the best balance between sharpness and realism.
- ResNet50 also performed well, but required more training time.

Model	MAE↓	PSNR↑	SSIM↑
U-Net + MSE Loss	5.5189	30.3006	0.9555
U-Net + Perceptual VGG16	6.0318	29.5798	0.9533
U-Net + Perceptual ResNet50	5.7462	29.9902	0.9513

3.3. QUALITATIVE COMPARISONS (1)

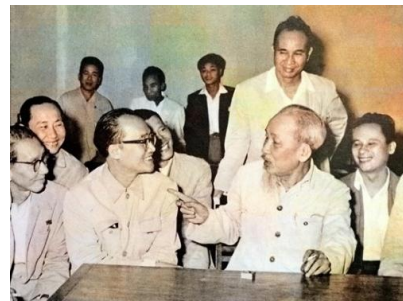


3.3. QUALITATIVE COMPARISONS (2)

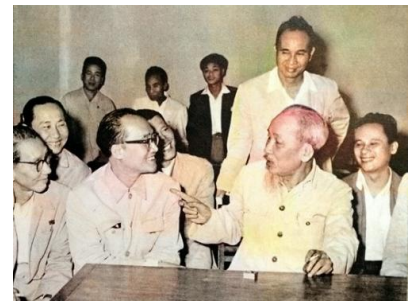
Grayscale



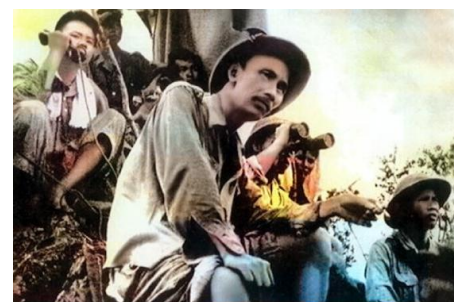
MSE



Perceptual
VGG



Perceptual
ResNet



Grayscale



MSE



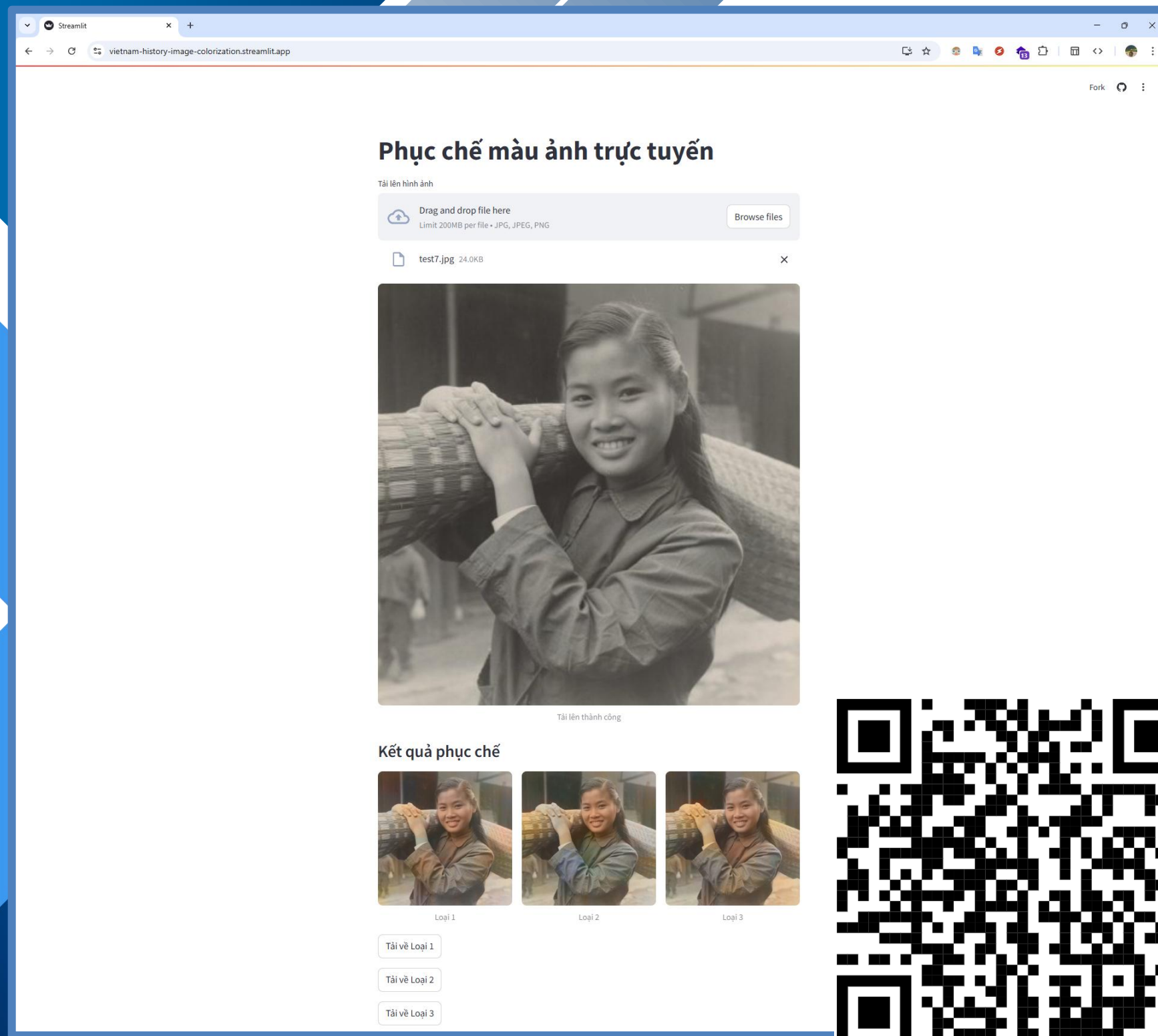
Perceptual
VGG



Perceptual
ResNet



3.4. APP DEPLOYMENT



- The model is deployed on Streamlit for public interaction.
- Users upload a grayscale photo and receive a colorized version in seconds.
- This makes historical restoration accessible to non-technical users



CHAPTER 4: CONCLUSION & FUTURE WORK



4.1. CONCLUSION



A U-Net with VGG16
perceptual loss performed
best in both PSNR and
quality

Deep learning can offer realistic
colorization for historical
Vietnamese photos.

Web deployment
increases the practical
impact of the research.

4.2. LIMITATION



01

The predicted colors are sometimes inaccurate due to task ambiguity.



02

The training dataset lacks diversity, limiting the model's generalization to true historical images.

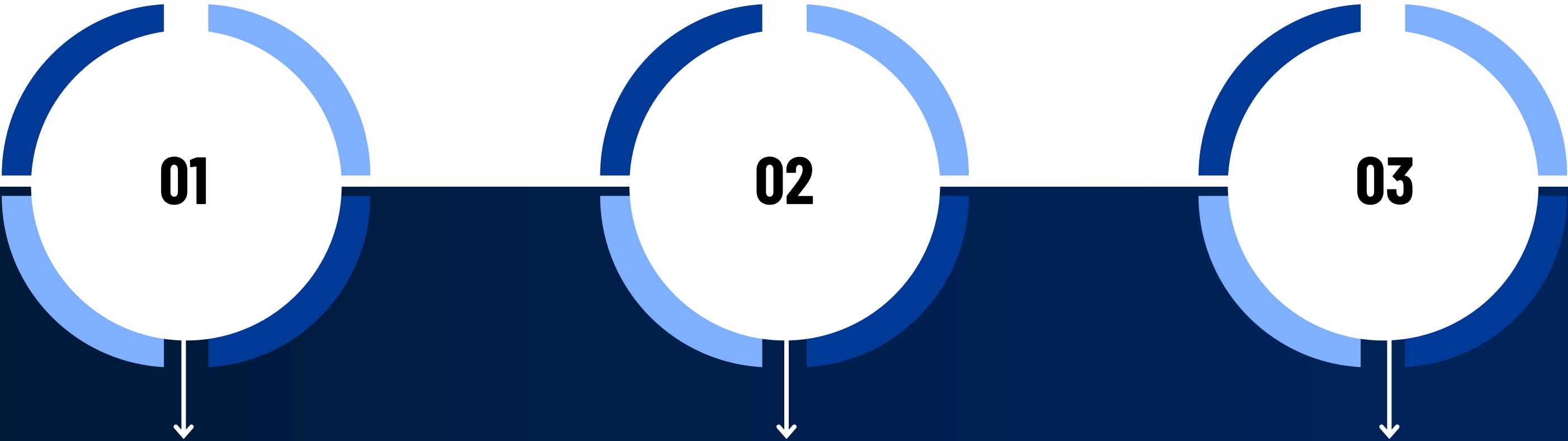


03

High training cost for perceptual models may limit deployment in resource-constrained environments.



4.3. FUTURE WORK



01

Incorporate GANs for more realistic and vibrant color outputs.

02

Include authentic historical Vietnamese photographs to improve generalization.

03

Use FastAPI and Streamlit to optimize response time and scalability.

THANK YOU

FOR YOUR ATTENTION AND
PARTICIPATION

